



Ultra Wide Band Low Noise Amplifier 0.01GHz~30GHz

Features

- Gain: 40dB Typical
- Noise Figure: 3.5dB Typical
- P1dB Output Power: +28dBm Typical
- PSAT Output Power: 30dBm
- Supply Voltage: +12V
- 50 Ohm Matched



Typical Applications

- Wireless Infrastructure
- 5G communication
- Test and measurement Instrument

RF Microwave & VSAT
Fiber Optics

Parameter	Min.	Typ.	Max.	Min.	Typ.	Max.	Units
Frequency Range	0.01		18	18		30	GHz
Gain	35	40		33	38		dB
Gain Flatness		± 2.5	± 3.5		± 1.0	± 2.0	dB
Gain Variation Over Temperature (-40°C~+85°C)		± 1.0			± 1.5		dB
*Noise Figure		3.0			4.5	5.5	dB
Input VSWR		1.5			1.7		: 1
Output VSWR		1.5			1.5		: 1
Output 1dB Compression Point (P1dB)	24	28		21	24		dBm
Saturated Output Power (Psat)		30			25		dBm
Output Third Order Intercept (OIP3)		37			33		dBm
Supply Current (Vcc=+12V)		550	700		550	700	mA
Isolation S12		-60			-55		dB

Weight	Net	4 Max.ounces	Impedance	50ohms
	Including Heat sink	6.2 Max. ounces		
Input / Output Connectors		2.92mm-Female	Material	Copper
Finish	Gold Plated	Package Sealing	Epoxy Sealed (Standard)	
			Hermetically Sealed (Option with extra charge)	

* Noise Figure at 0.01-0.1GHz is 6dB Typical

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Absolute Maximum Ratings

Operating Voltage	+15V
RF Input Power (RFIN)	-3dBm

Biasing Up Procedure

Step 1	Connect Ground Pin
Step 2	Connect input and output
Step 3	Connect +12V biasing

Power OFF Procedure

Step 1	Turn off +12V biasing
Step 2	Remove RF connection
Step 3	Remove Ground

Outline Drawing:

All Dimensions in mm (inches)

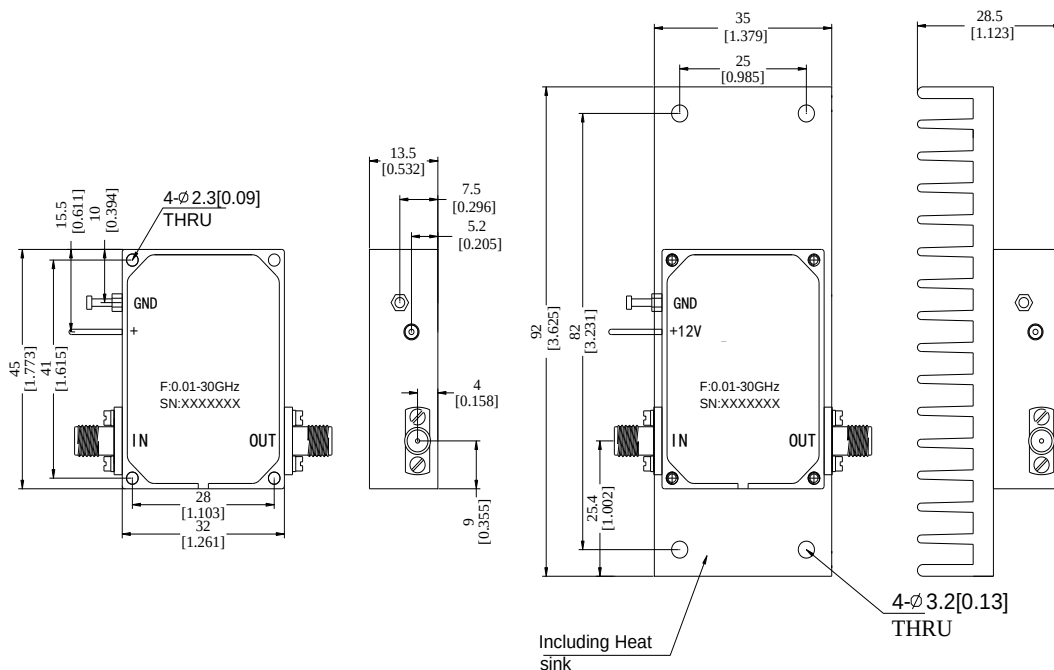
Housing Tolerances ± 0.1 (0.004)

(Excl Heat Sink).

Environmental Specifications

Operational Temperature	-40°C~+85°C(Case Temperature)
Storage Temperature	-50°C~+105°C
Altitude	30,000 ft. (Epoxy Sealed Controlled environment)
	60,000 ft. 1.0psi min (Hermetically Sealed Un-controlled environment) (Optional)
Vibration	25g RMS (15 degrees 2KHz) endurance, 1 hour per axis
Humidity	100% RH at 35°C, 95%RH at 40°C
Shock	20G for 11msec half sine wave, 3 axis both directions

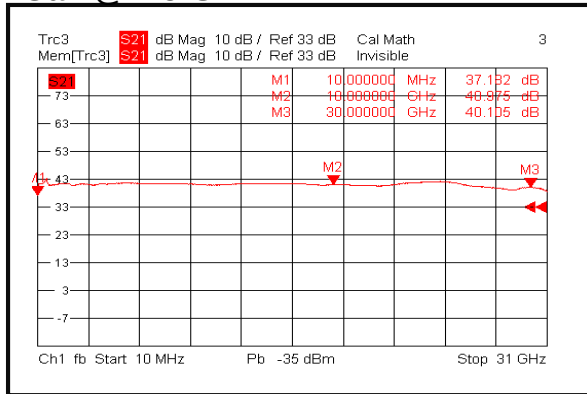
Heat Sink required during operation(Sold Separately)



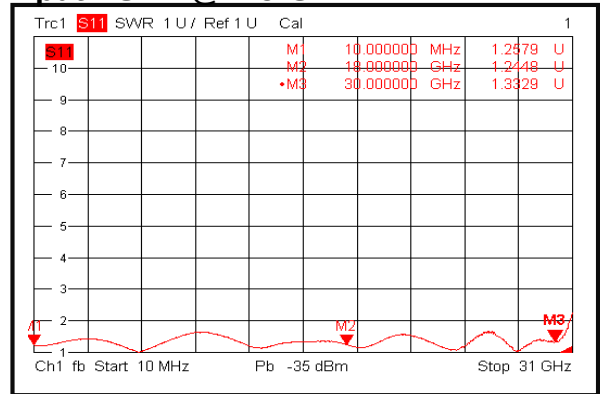


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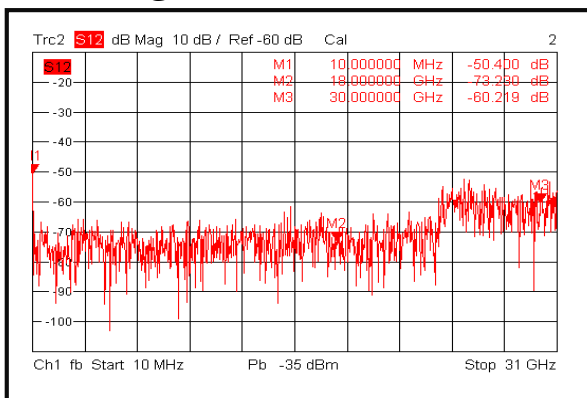
Gain@+25°C



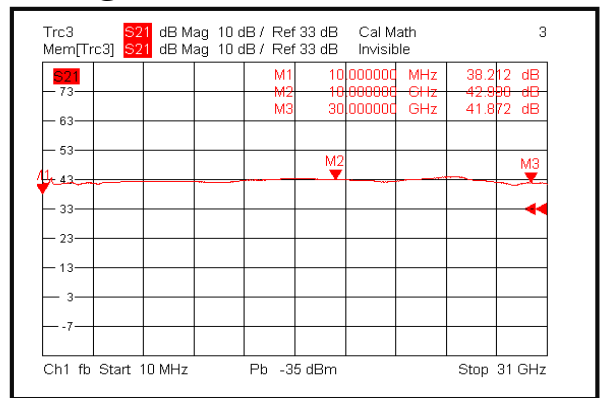
Input VSWR@+25°C



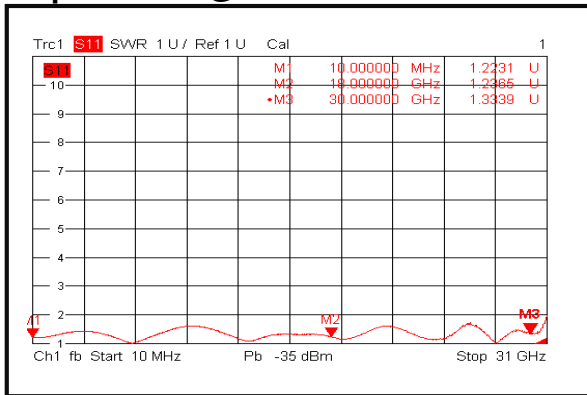
Isolation@+25°C



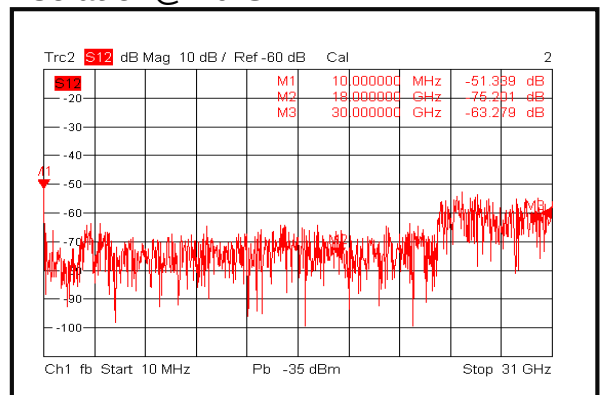
Gain@-40°C



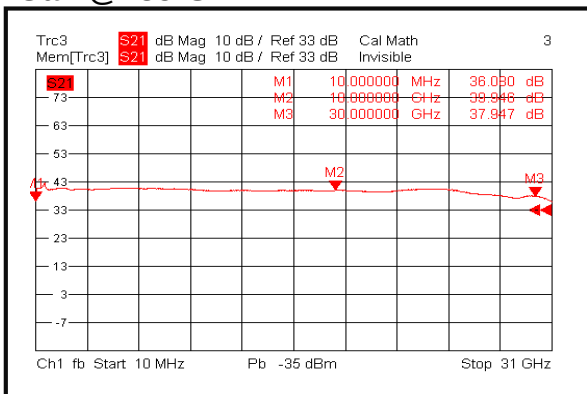
Input VSWR@-40°C



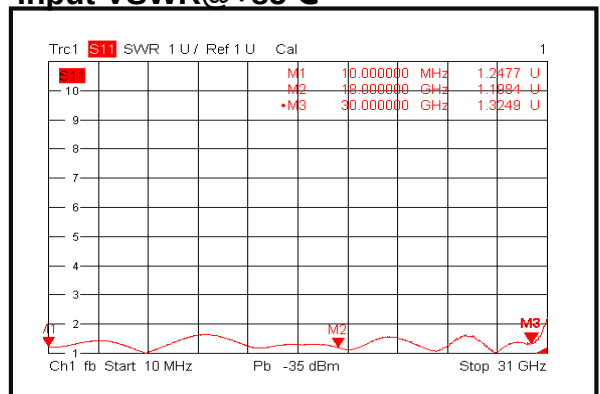
Isolation@-40°C



Gain@+85°C



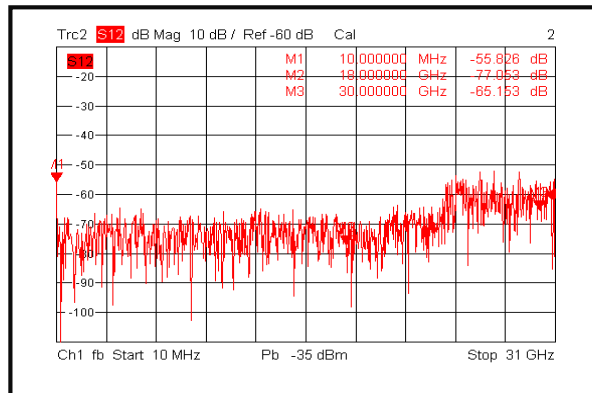
Input VSWR@+85°C



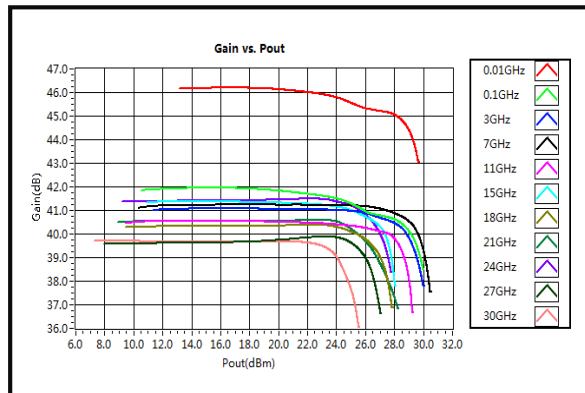


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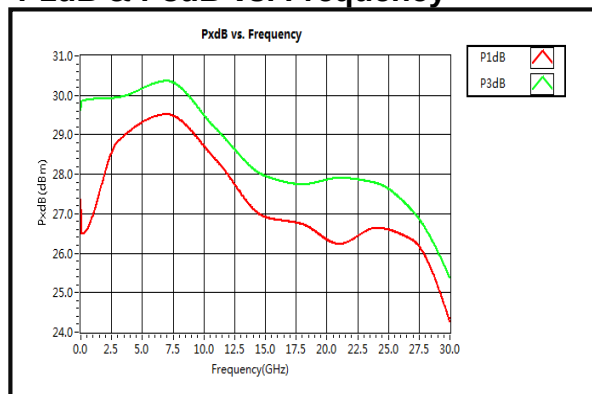
Isolation@+85°C



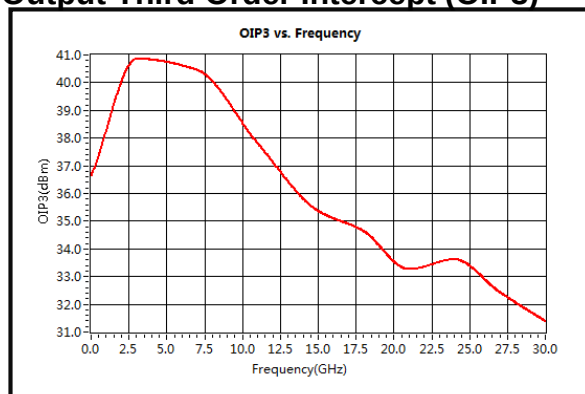
Gain vs. Output Power



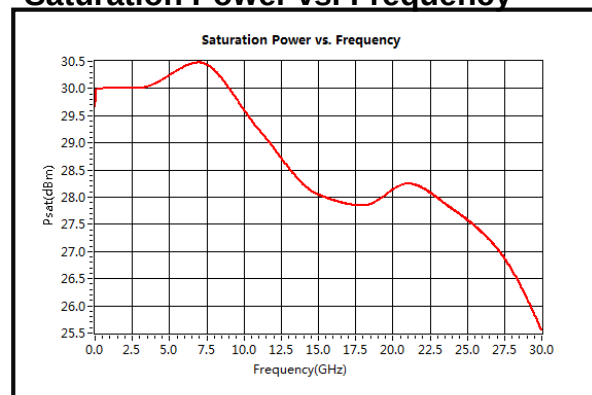
P1dB & P3dB vs. Frequency



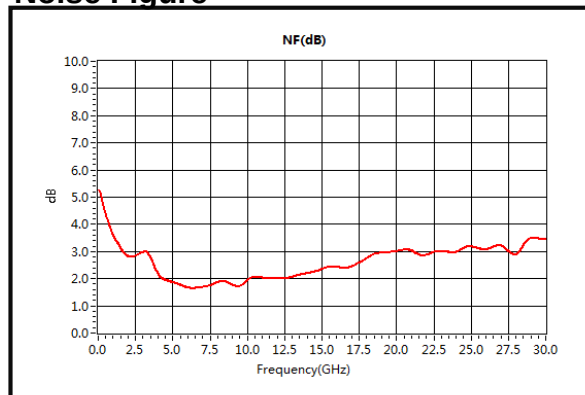
Output Third Order Intercept (OIP3)



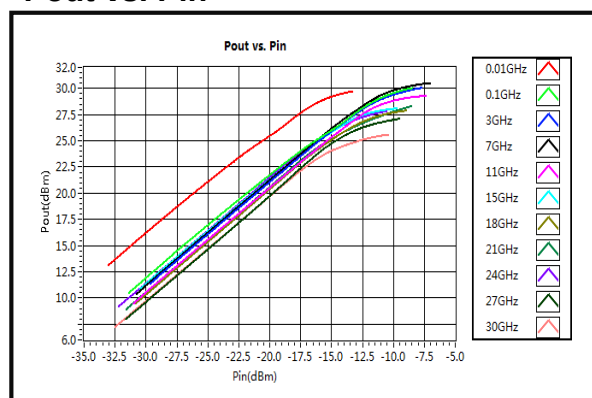
Saturation Power vs. Frequency



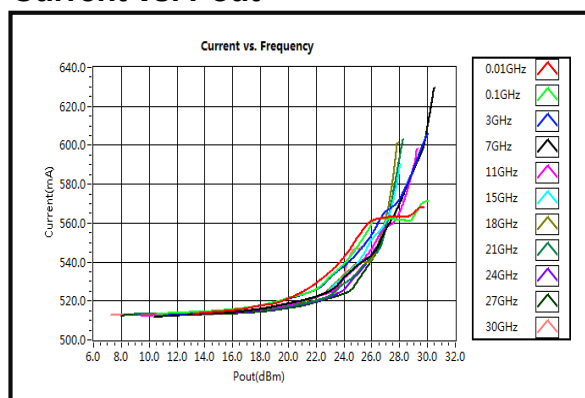
Noise Figure



Pout vs. Pin



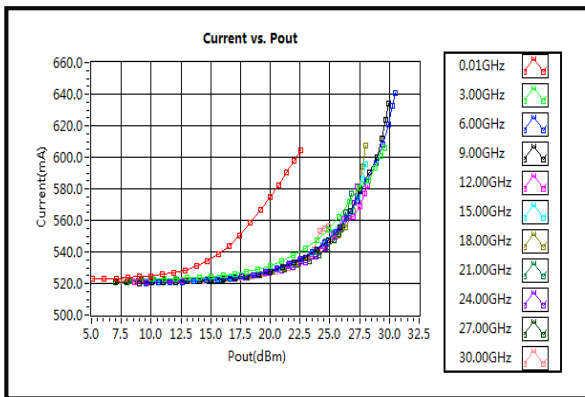
Current vs. Pout



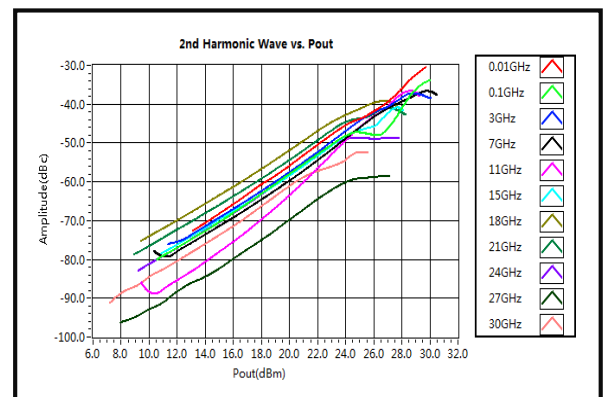


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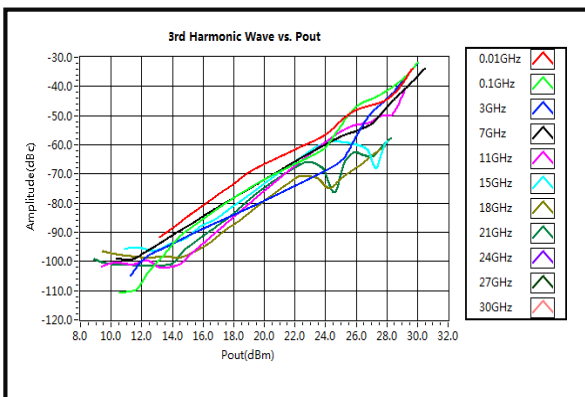
Current vs. Pout



2nd Harmonic Wave Output Power



3rd Harmonic Wave Output Power



4th Harmonic Wave Output Power

